

OSSP (Operating Systems And Systems Programming) - 25CS2104E

1) Develop a C program that demonstrates how a Linux operating system executes a command entered by a user that 1. Accept a Linux command as input. 2. Create a child process using `fork()`. 3. Execute the command in the child process using an appropriate `exec()` system call. 4. Allow the parent process to wait for the child using `wait ()`. 5. Display the Process ID (PID) of both parent and child processes.

Using Linux terminal commands (`uname`, `lscpu`, `lsblk`, `ps`, `top`), investigate the relationship between hardware resources and operating system services. Prepare a report explaining how the OS abstracts CPU, memory, storage, and I/O devices

```
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>

int main() {
    char cmd[100];

    // Prompt user to enter a standard Linux terminal command
    printf("Enter Linux command: ");
    scanf("%99s", cmd);

    // Create a new process by duplicating the calling process
    pid_t pid = fork();

    if (pid == 0) {
        // --- CHILD PROCESS EXECUTION BLOCK ---
        // getpid() returns the Process ID assigned to the child
        printf("Child PID: %d\n", getpid());

        // Replace the child process image with the specified command.
        // execlp searches the PATH environment variable for the binary.
        execlp(cmd, cmd, NULL);

        // Executed only if execlp fails (e.g., command not found)
        perror("Execution Failed");
        exit(1);
    }
    else if (pid > 0) {
        // --- PARENT PROCESS EXECUTION BLOCK ---
        // Display the PID of the running parent process
        printf("Parent PID: %d\n", getpid());

        // Suspend parent execution until the child process terminates
        wait(NULL);

        printf("Child process completed.\n");
    }
    else {
        // --- ERROR HANDLING BLOCK ---
        // If fork() returns a negative number, process creation failed
        printf("Fork Failed\n");
        return 1;
    }

    return 0;
}
```

```
worker -c MDSImporterWorker -m com.apple.mdworker.shared
501 32041 1 0 10:31PM ?? 0:00.13 /System/Library/Frameworks/CoreServices.framework/Versions/A/Frameworks/M
worker -c MDSImporterWorker -m com.apple.mdworker.shared
0 30078 30015 0 10:35AM ttys000 0:00.02 login -pf medikondarakesh
501 30079 30078 0 10:35AM ttys000 0:00.06 -zsh
0 30154 30015 0 10:59AM ttys001 0:00.01 login -pf medikondarakesh
501 30155 30154 0 10:59AM ttys001 0:00.04 -zsh
0 30237 30015 0 11:13AM ttys002 0:00.02 login -pf medikondarakesh
501 30238 30237 0 11:13AM ttys002 0:00.22 -zsh
0 31984 30015 0 10:23PM ttys003 0:00.02 login -pf medikondarakesh
501 31985 31984 0 10:23PM ttys003 0:00.04 -zsh
0 32042 31985 0 10:31PM ttys003 0:00.01 ps -ef
medikondarakesh@Medikondas-MacBook-Air ~ %
```

```
[medikondarakesh@Medikondas-MacBook-Air ~ % nano task1.c
[medikondarakesh@Medikondas-MacBook-Air ~ % gcc task1.c -o task1
[medikondarakesh@Medikondas-MacBook-Air ~ % ./task1
Enter Linux command: uname -a
Parent PID: 32001
Child PID: 32002
Darwin
Child process completed.
medikondarakesh@Medikondas-MacBook-Air ~ % █
```

```
[medikondarakesh@Medikondas-MacBook-Air ~ % diskutil list
/dev/disk0 (internal, physical):
#:

| #: | TYPE                          | NAME  | SIZE      | IDENTIFIER |
|----|-------------------------------|-------|-----------|------------|
| 0: | GUID_partition_scheme         | -     | *251.0 GB | disk0      |
| 1: | Apple_APFS_ISC Container      | disk1 | 524.3 MB  | disk0s1    |
| 2: | Apple_APFS Container          | disk2 | 245.1 GB  | disk0s2    |
| 3: | Apple_APFS_Recovery Container | disk3 | 5.4 GB    | disk0s3    |


/dev/disk2 (synthesized):
#:

| #: | TYPE                  | NAME                    | SIZE      | IDENTIFIER |
|----|-----------------------|-------------------------|-----------|------------|
| 0: | APFS Container Scheme | -                       | +245.1 GB | disk2      |
|    |                       | Physical Store disk0s2  |           |            |
| 1: | APFS Volume           | Macintosh HD - Data     | 140.5 GB  | disk2s1    |
| 2: | APFS Volume           | Macintosh HD            | 12.6 GB   | disk2s3    |
| 3: | APFS Snapshot         | com.apple.os.update-... | 12.6 GB   | disk2s3s1  |
| 4: | APFS Volume           | Preboot                 | 9.0 GB    | disk2s4    |
| 5: | APFS Volume           | Recovery                | 1.3 GB    | disk2s5    |
| 6: | APFS Volume           | VM                      | 24.6 KB   | disk2s6    |


/dev/disk4 (disk image):
#:

| #: | TYPE                  | NAME              | SIZE    | IDENTIFIER |
|----|-----------------------|-------------------|---------|------------|
| 0: | GUID_partition_scheme | -                 | +1.6 GB | disk4      |
| 1: | Apple_HFS             | ChatGPT Installer | 1.5 GB  | disk4s1    |


/dev/disk5 (disk image):
#:

| #: | TYPE                  | NAME               | SIZE      | IDENTIFIER |
|----|-----------------------|--------------------|-----------|------------|
| 0: | GUID_partition_scheme | -                  | +625.4 MB | disk5      |
| 1: | Apple_HFS             | Arduino IDE 2.3.10 | 625.3 MB  | disk5s1    |


/dev/disk6 (disk image):
#:

| #: | TYPE | NAME                   | SIZE    | IDENTIFIER |
|----|------|------------------------|---------|------------|
| 0: |      | MySQL Workbench com... | +1.3 GB | disk6      |


/dev/disk7 (disk image):
#:

| #: | TYPE                  | NAME                    | SIZE      | IDENTIFIER |
|----|-----------------------|-------------------------|-----------|------------|
| 0: | GUID_partition_scheme | -                       | +629.2 MB | disk7      |
| 1: | Apple_HFS             | mysql-9.7.1-macos15-... | 629.2 MB  | disk7s1    |


/dev/disk8 (disk image):
#:

| #: | TYPE                  | NAME    | SIZE      | IDENTIFIER |
|----|-----------------------|---------|-----------|------------|
| 0: | GUID_partition_scheme | -       | +996.9 MB | disk8      |
| 1: | Apple_HFS             | VS Code | 996.9 MB  | disk8s1    |


medikondarakesh@Medikondas-MacBook-Air ~ % █
```

Use the strace utility to trace all system calls generated by the command `cat sample.txt`. Analyze the sequence of system calls and identify the kernel services involved.

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```
-- -zsh -- -zsh -- -zsh
```

```
501 32034      1 0 10:31PM ??           0:00.22 /System/Library/Frameworks/CoreServices.framework/Versions/A/Frameworks/Metadata.framework/Versions/A/Support/mdworker_shared -s md
worker -c MDSImporterWorker -m com.apple.mdworker.shared
501 32035      1 0 10:31PM ??           0:00.11 /System/Library/Frameworks/CoreServices.framework/Versions/A/Frameworks/Metadata.framework/Versions/A/Support/mdworker_shared -s md
worker -c MDSImporterWorker -m com.apple.mdworker.shared
501 32036      1 0 10:31PM ??           0:00.07 /System/Library/Frameworks/CoreServices.framework/Versions/A/Frameworks/Metadata.framework/Versions/A/Support/mdworker_shared -s md
worker -c MDSImporterWorker -m com.apple.mdworker.shared
501 32039      1 0 10:31PM ??           0:00.08 /System/Library/Frameworks/CoreServices.framework/Versions/A/Frameworks/Metadata.framework/Versions/A/Support/mdworker_shared -s md
worker -c MDSImporterWorker -m com.apple.mdworker.shared
501 32041      1 0 10:31PM ??           0:00.13 /System/Library/Frameworks/CoreServices.framework/Versions/A/Frameworks/Metadata.framework/Versions/A/Support/mdworker_shared -s md
worker -c MDSImporterWorker -m com.apple.mdworker.shared
0 30078 30015 0 10:35AM ttys000       0:00.02 login -pf medikondarakesh
501 30079 30078 0 10:35AM ttys000       0:00.06 -zsh
0 30154 30015 0 10:59AM ttys001       0:00.01 login -pf medikondarakesh
501 30155 30154 0 10:59AM ttys001       0:00.04 -zsh
0 30237 30015 0 11:13AM ttys002       0:00.02 login -pf medikondarakesh
501 30238 30237 0 11:13AM ttys002       0:00.22 -zsh
0 31984 30015 0 10:23PM ttys003       0:00.02 login -pf medikondarakesh
501 31985 31984 0 10:23PM ttys003       0:00.04 -zsh
0 32042 31985 0 10:31PM ttys003       0:00.01 ps -ef
medikondarakesh@Medikondas-MacBook-Air ~ % top
medikondarakesh@Medikondas-MacBook-Air ~ % nano task2.c
medikondarakesh@Medikondas-MacBook-Air ~ % echo "Operating System Systems Programming Lab" > input.txt
medikondarakesh@Medikondas-MacBook-Air ~ % gcc task2.c -o task2
medikondarakesh@Medikondas-MacBook-Air ~ % ./task2
File copied successfully.
File copied successfully.
copied: cannot open `copied' (No such file or directory)
successfully.: cannot open `successfully.' (No such file or directory)
medikondarakesh@Medikondas-MacBook-Air ~ % cat output.txt
Operating System Systems Programming Lab
Operating System Systems Programming Lab
zsh: command not found: Operating
medikondarakesh@Medikondas-MacBook-Air ~ % strace cat input.txt
zsh: command not found: strace
medikondarakesh@Medikondas-MacBook-Air ~ % sudo dtruss cat input.txt
Password:
Sorry, try again.
Password:
dtrace: system integrity protection is on, some features will not be available

dtrace: failed to execute cat: Operation not permitted
medikondarakesh@Medikondas-MacBook-Air ~ % sudo dtruss ./task2
dtrace: system integrity protection is on, some features will not be available

SYSCALL(args) = return
File copied successfully.

medikondarakesh@Medikondas-MacBook-Air ~ % sudo dtrace -n 'syscall:::entry /execname == "task2"/ { @[probefunc] = count(); }' -c ./task2
dtrace: system integrity protection is on, some features will not be available

dtrace: invalid probe specifier syscall:::entry /execname == "task2"/ { @[probefunc] = count(); }: probe description syscall:::entry does not match any probes. System Integrity Protection is on
medikondarakesh@Medikondas-MacBook-Air ~ %
```